

Non-Intrusive Load Monitoring (NILM) aims to identify when individual appliances are active using only aggregate household electricity consumption. Recent work such as CamAL has shown that appliance usage can be localized with weak supervision by training time-series classifiers using only coarse appliance-level labels and deriving timestamp-level predictions from Class Activation Maps (CAMs). This substantially reduces the need for expensive appliance-level annotations. However, CamAL primarily learns to distinguish whether an appliance is present within a time window, while localization is obtained indirectly from classifier explanations. As a result, accurately identifying the temporal boundaries and duration of appliance activations remains challenging, particularly for appliances with complex or overlapping consumption patterns.

This research investigates **Weakly Supervised Temporal Action Localization (WTAL)** as a framework for improving appliance localization from coarse labels. Instead of treating localization only as a post-hoc explanation of a classifier, the proposed approach aims to explicitly learn temporally discriminative representations that identify the most relevant segments of an aggregate power sequence and refine the start and end boundaries of appliance activations. The framework will explore temporal attention and localization mechanisms while retaining CamAL's weak-supervision setting, where only appliance presence or possession information is required during training. The approach will be evaluated on real-world NILM datasets using localization metrics such as F1-score, precision, recall, and matching ratio, with CamAL and existing weakly and strongly supervised NILM methods used as baselines. The objective is to achieve **more precise appliance localization without sacrificing the label efficiency that makes weakly supervised NILM practical and scalable.**